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Seo et al.

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(54) **RADAR ANTENNA**

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G01S 7/03 (2006.01)
H01Q 1/52 (2006.01)

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CPC ***H01Q 13/106*** (2013.01); ***G01S 7/03*** (2013.01); ***H01Q 1/526*** (2013.01)

(58) **Field of Classification Search**
CPC H01Q 13/106; H01Q 1/526; H01Q 1/521; H01Q 21/064; H01Q 1/3233; G01S 7/03; G01S 7/038; G01S 13/931
See application file for complete search history.

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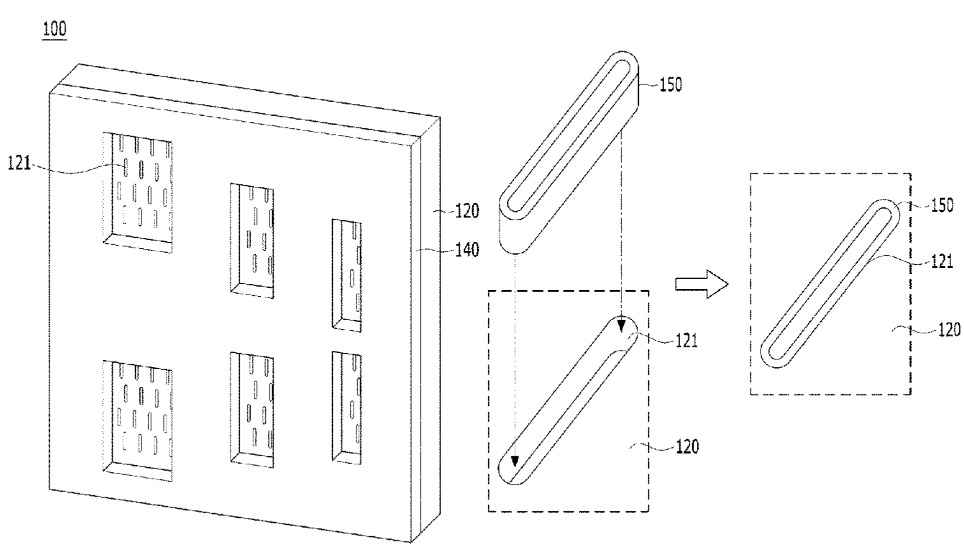
(74) *Attorney, Agent, or Firm* — Maschoff Brennan

(57) **ABSTRACT**

Disclosed is a radar antenna which has a shielding space corresponding to each antenna of an antenna body by using an accommodation hole of a shielding member to prevent mutual coupling between antennas. The disclosed radar antenna comprises an antenna body which has a first surface and a second surface and in which a plurality first slot groups are formed to be spaced apart from each other on the first surface, and a shielding member which is stacked on the first surface of the antenna body and in which a plurality of accommodation holes are formed to respectively overlap the plurality of first slot groups.

13 Claims, 16 Drawing Sheets



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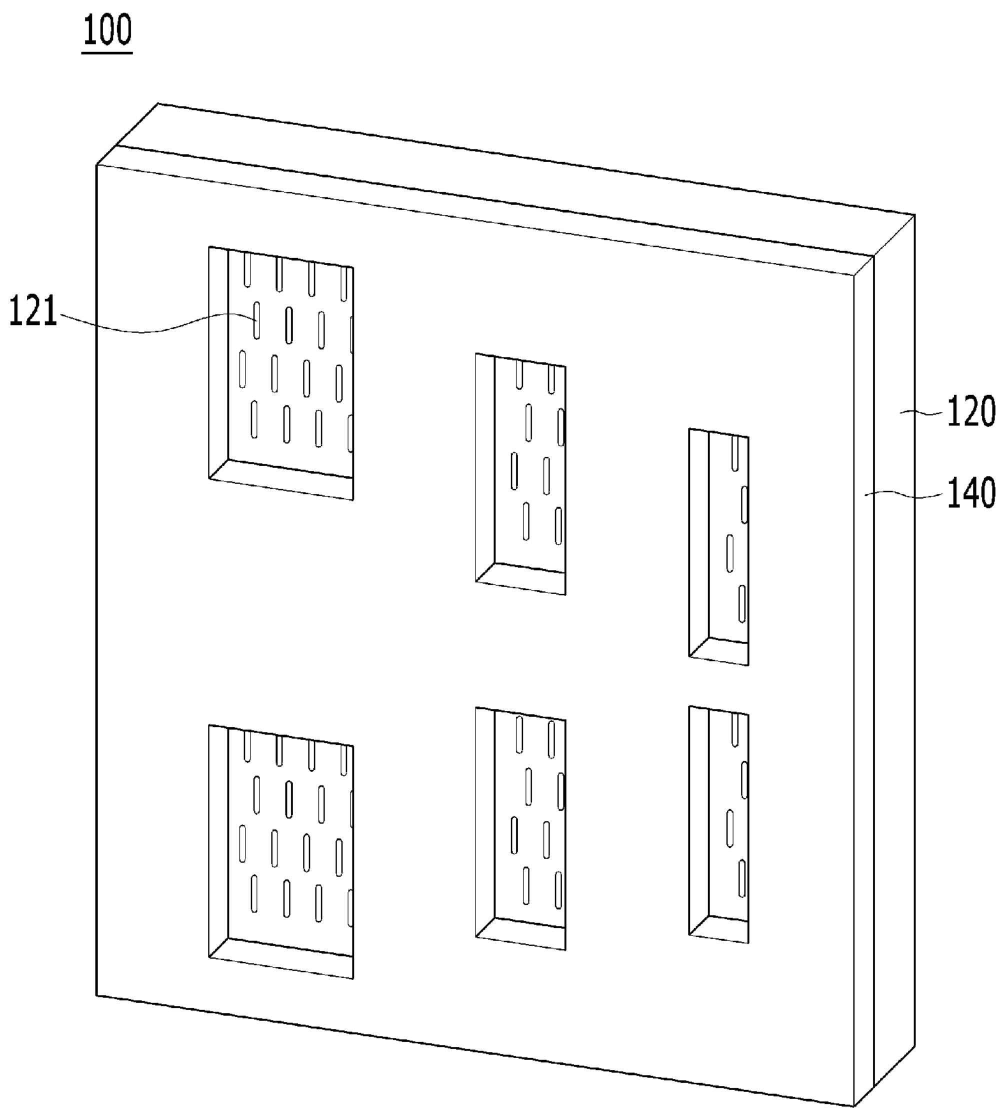
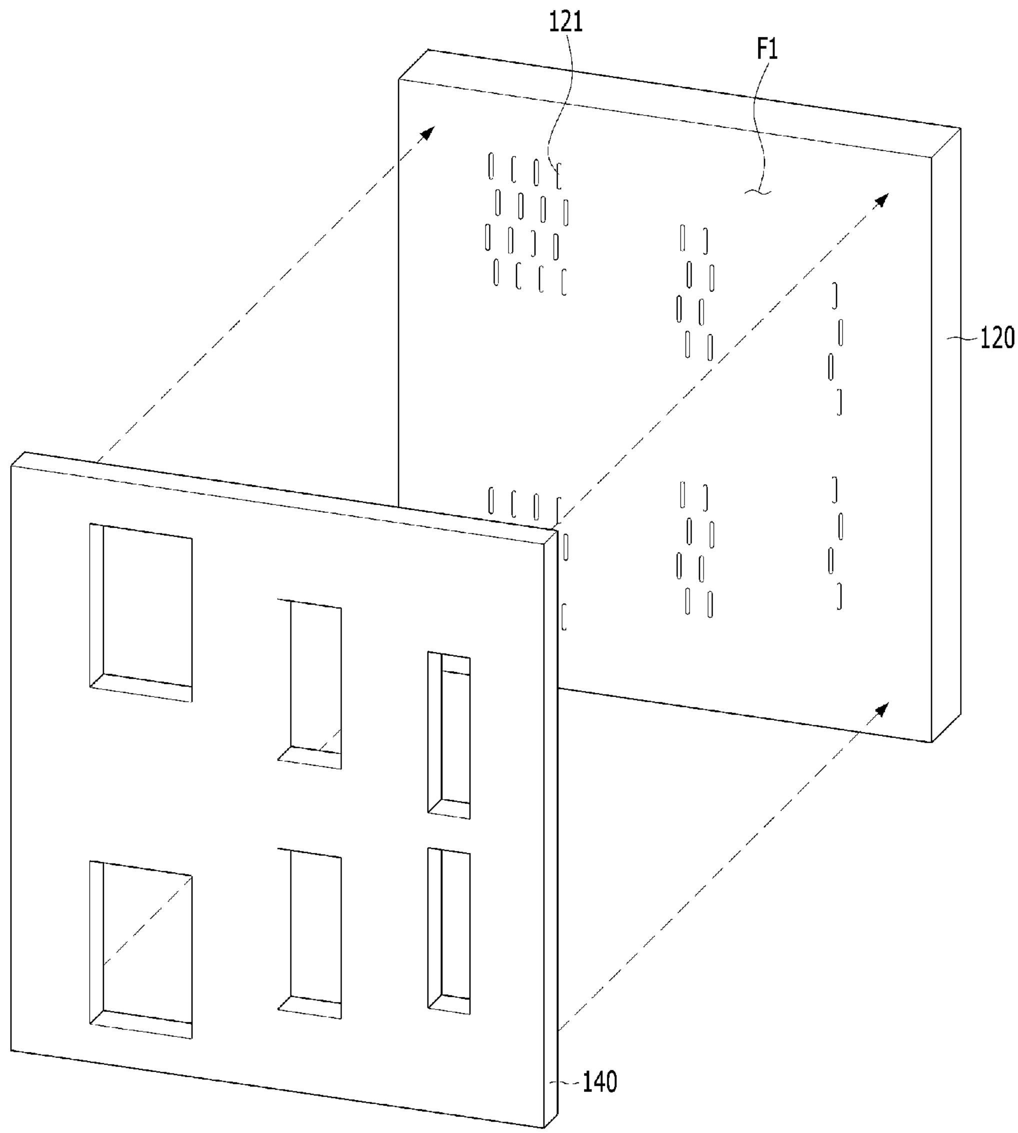
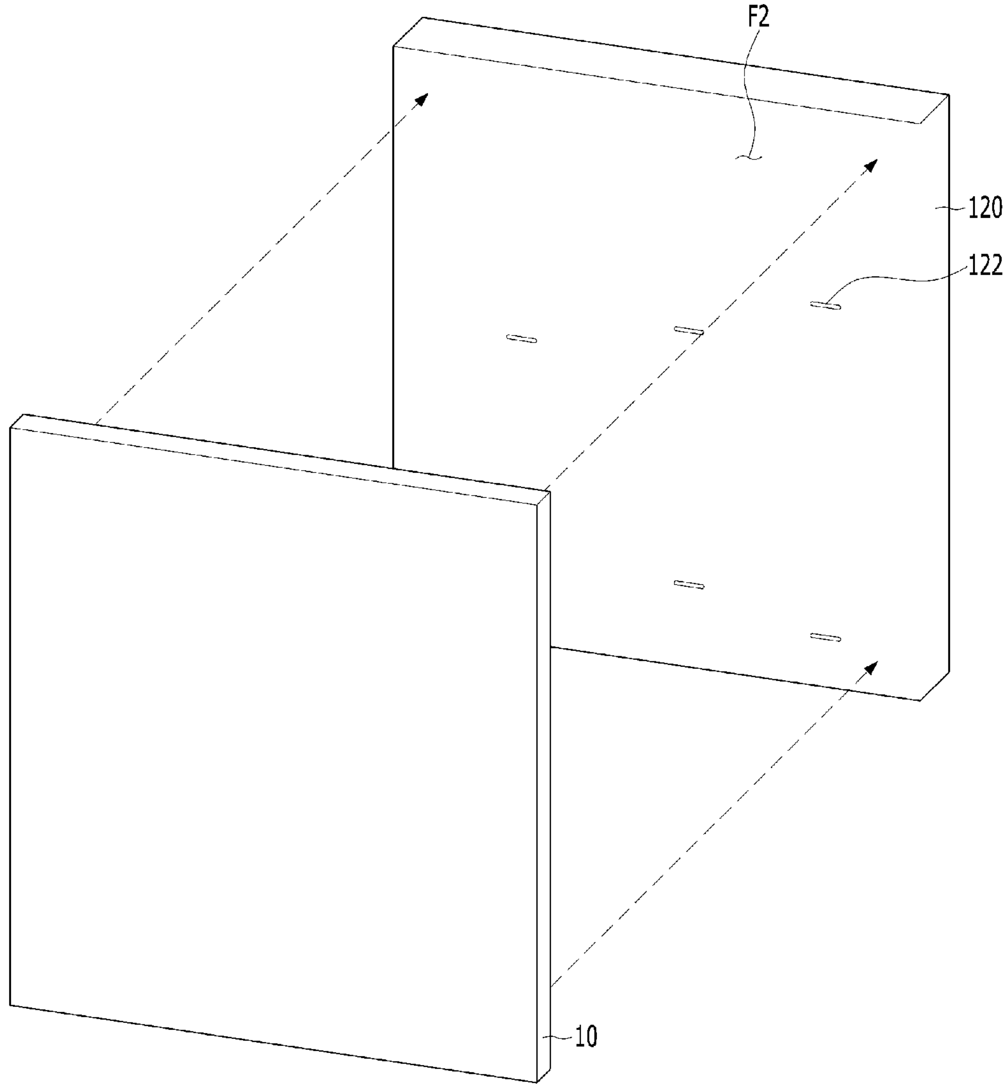
[FIG. 1]



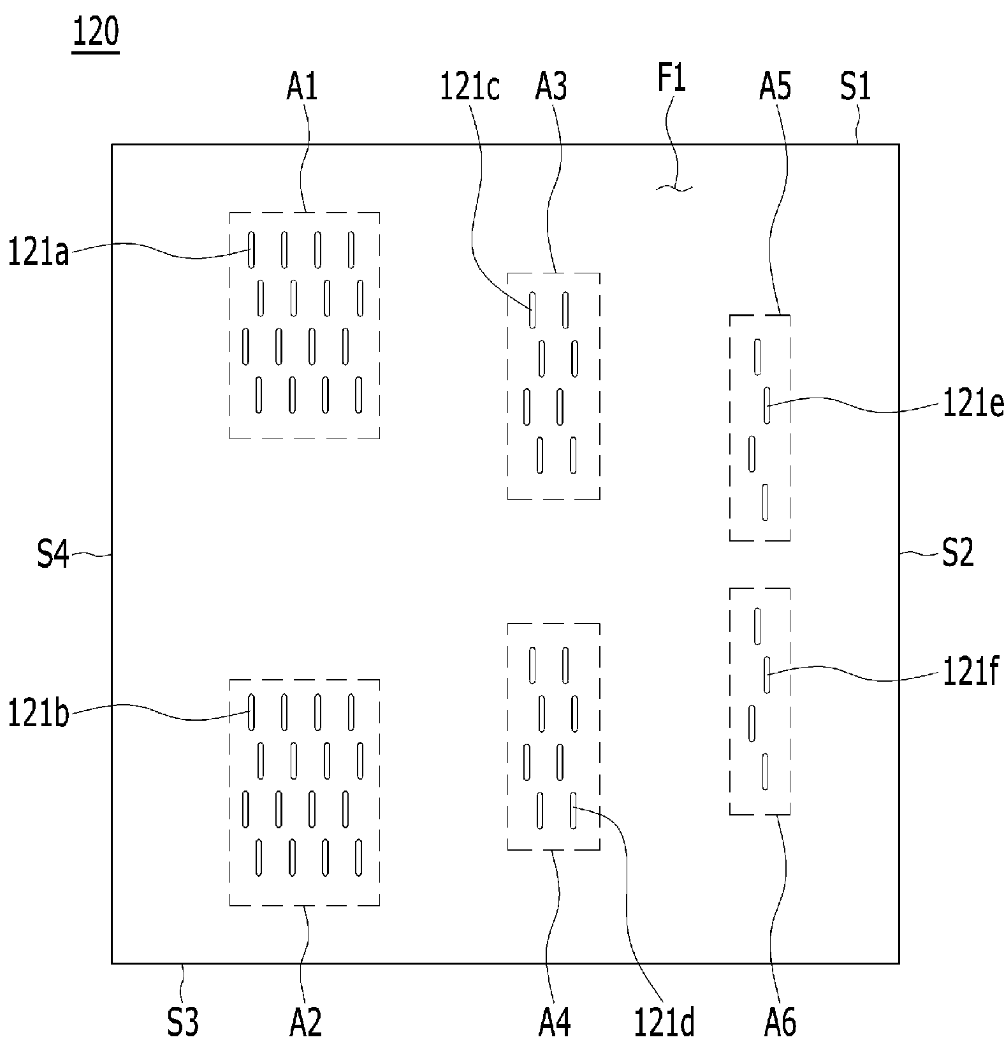
[FIG. 2]



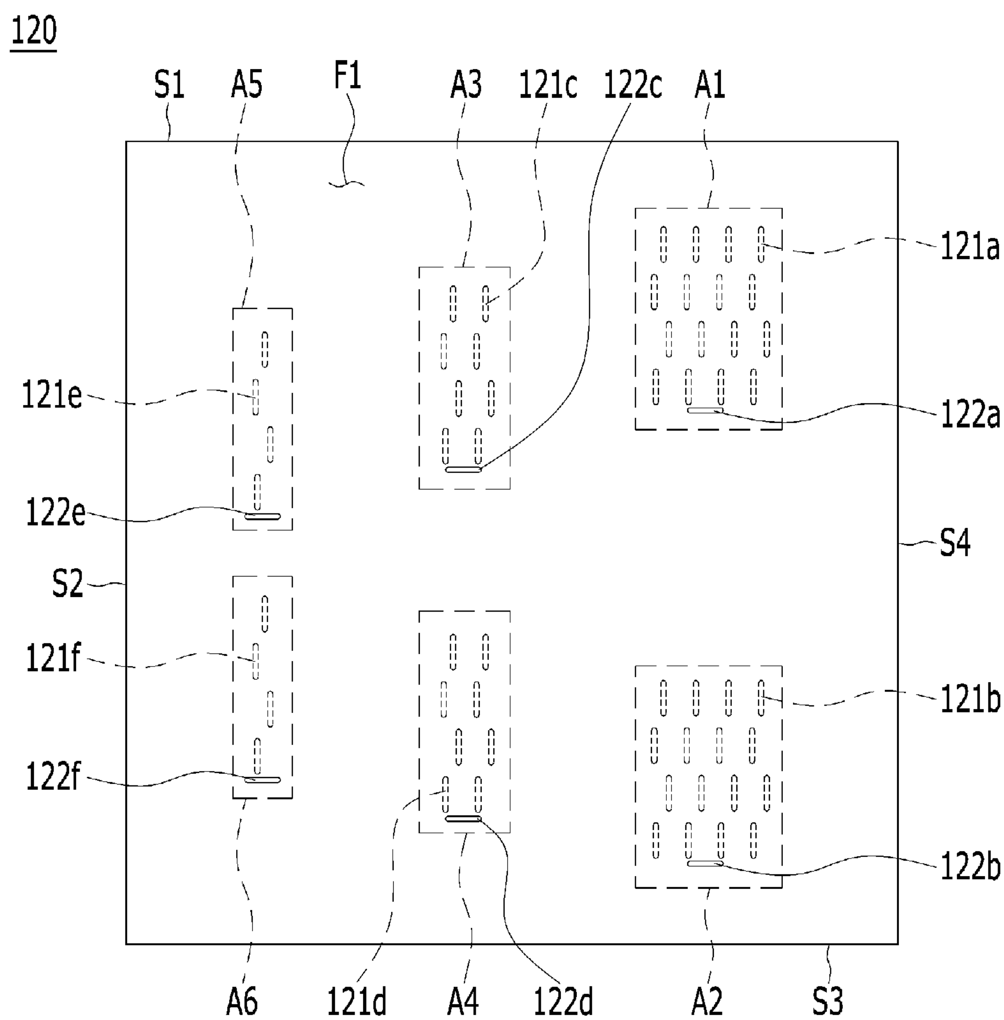
[FIG. 3]



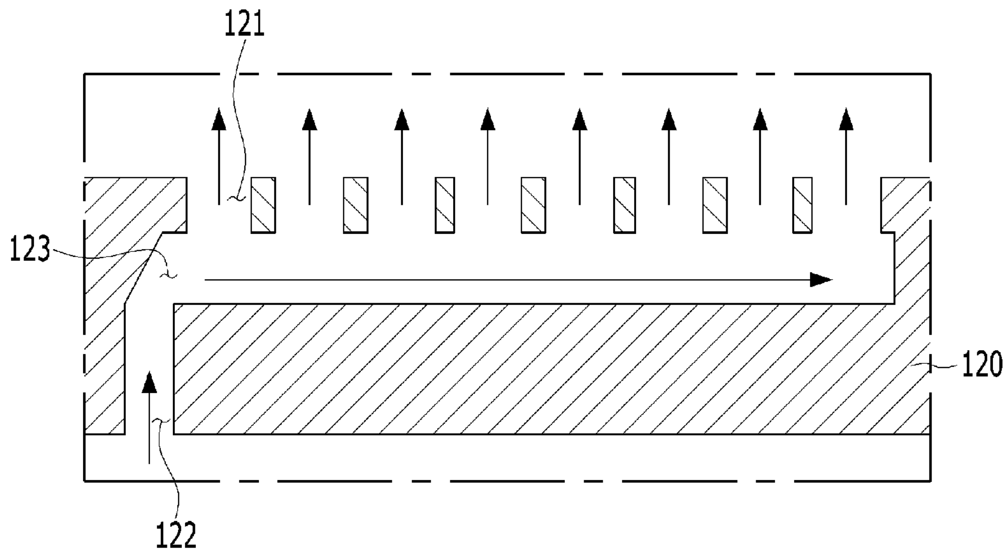
[FIG. 4]



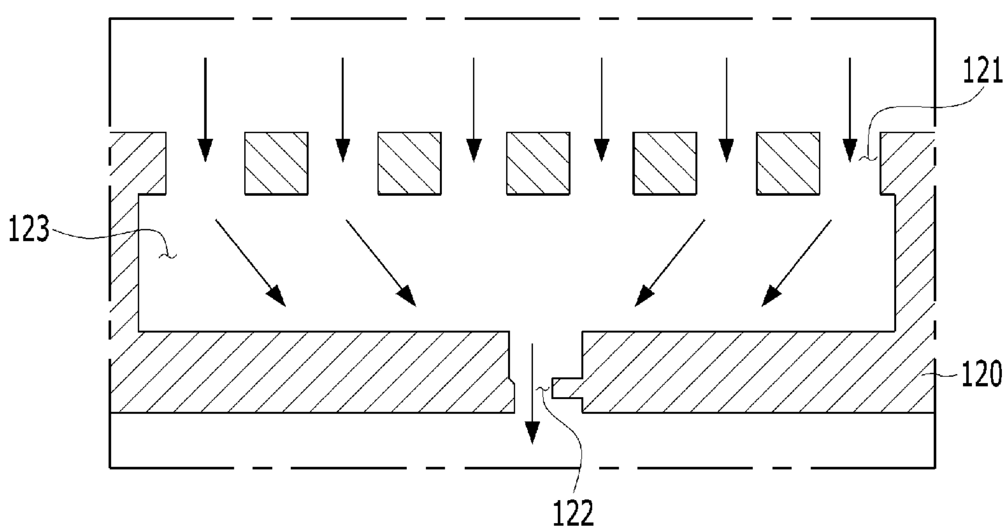
[FIG. 5]


120
S1
A5
F1
A3
121c
122c
A1
121a
122a
121e
122e
S2
121f
122f
S4
121b
122b
A6
121d
A4
122d
A2
S3

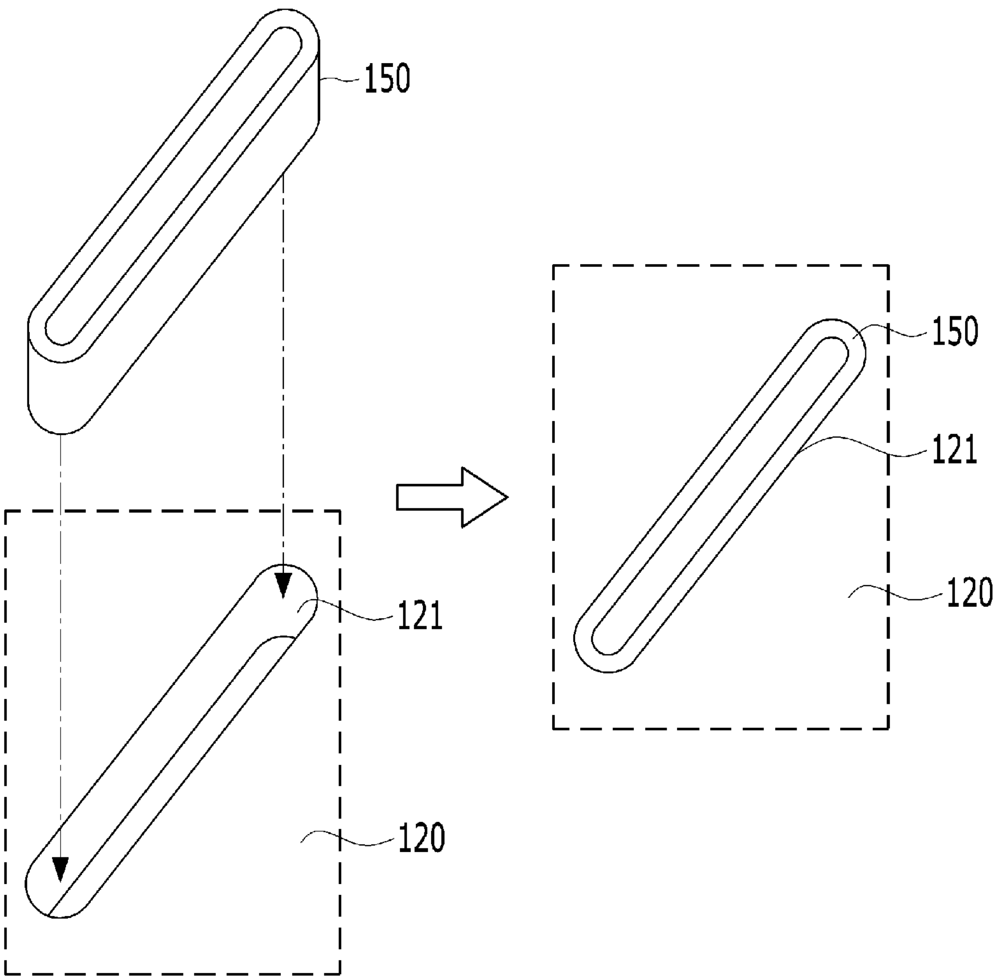
[FIG. 6]



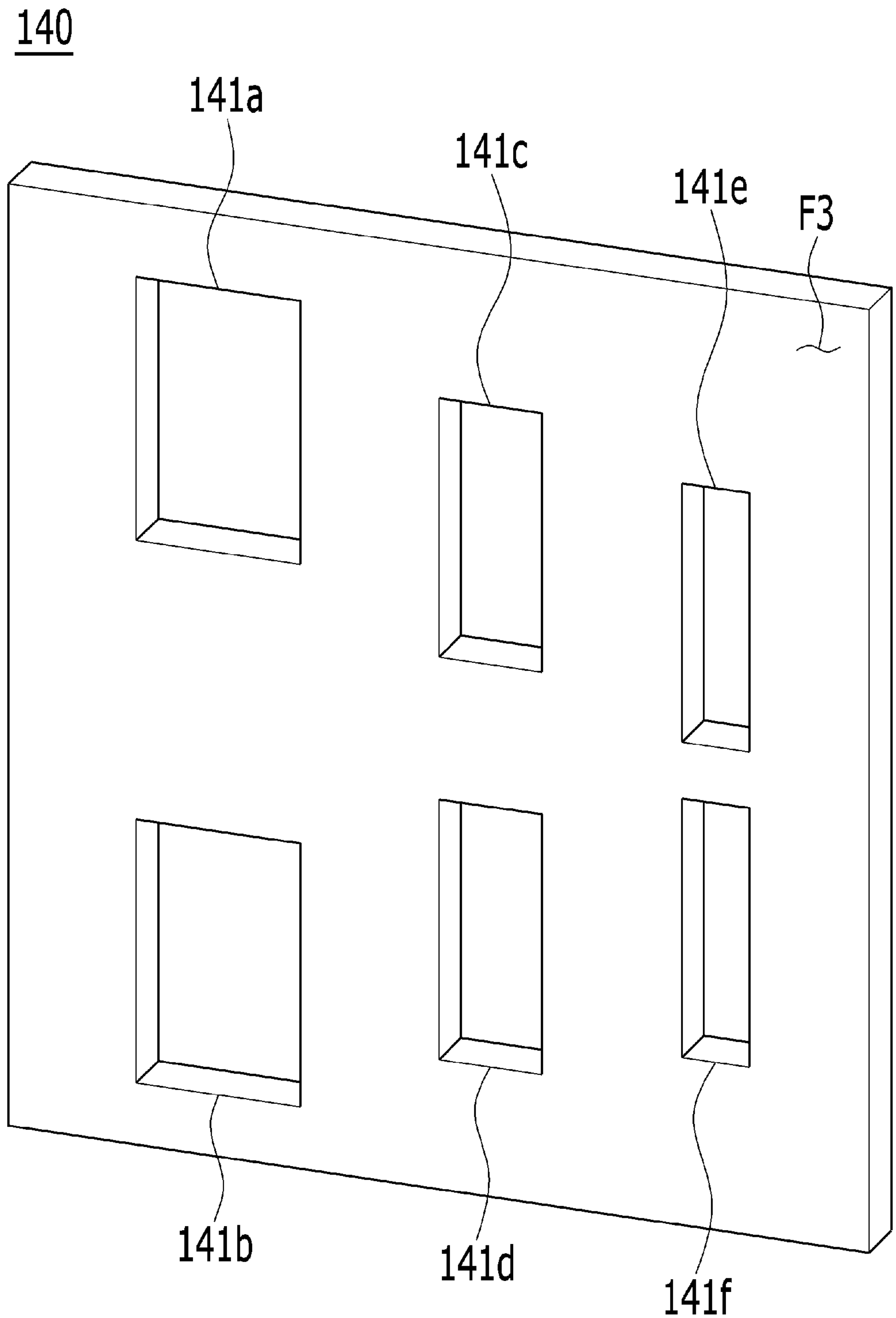
[FIG. 7]



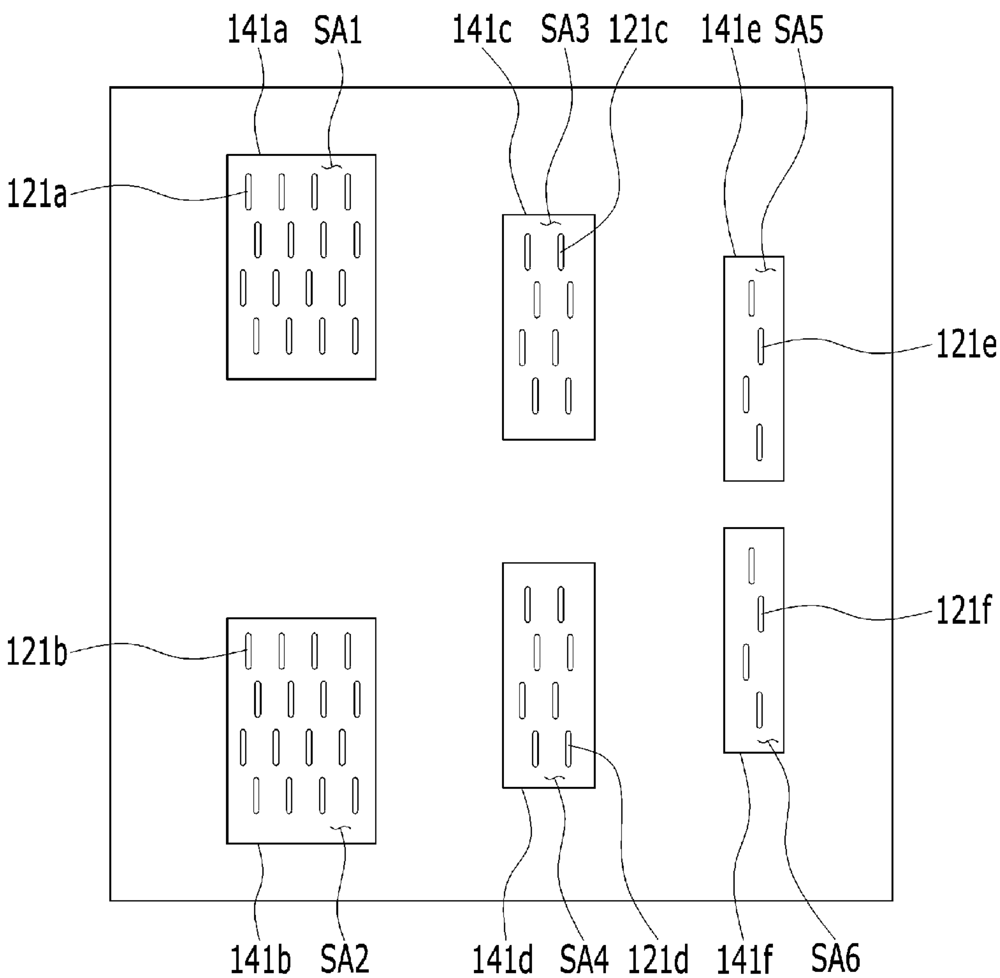
[FIG. 8]



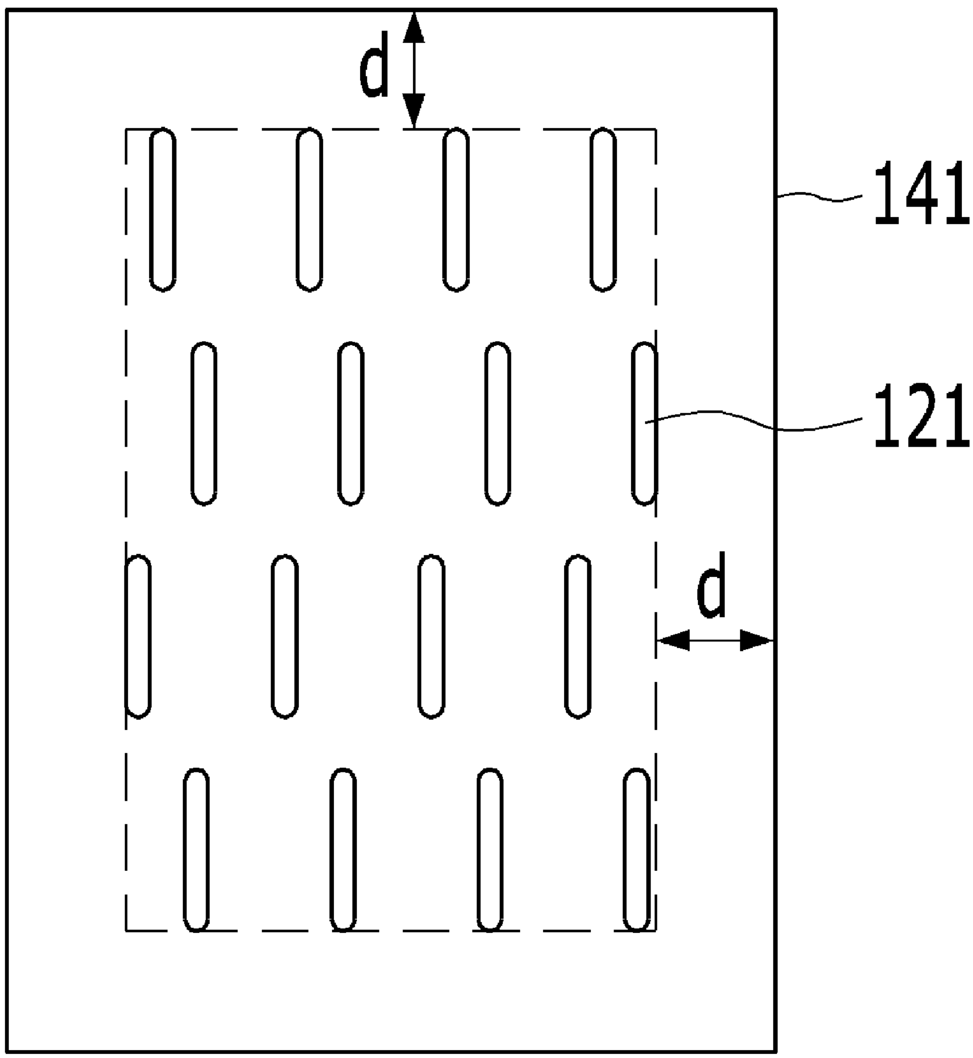
[FIG. 9]



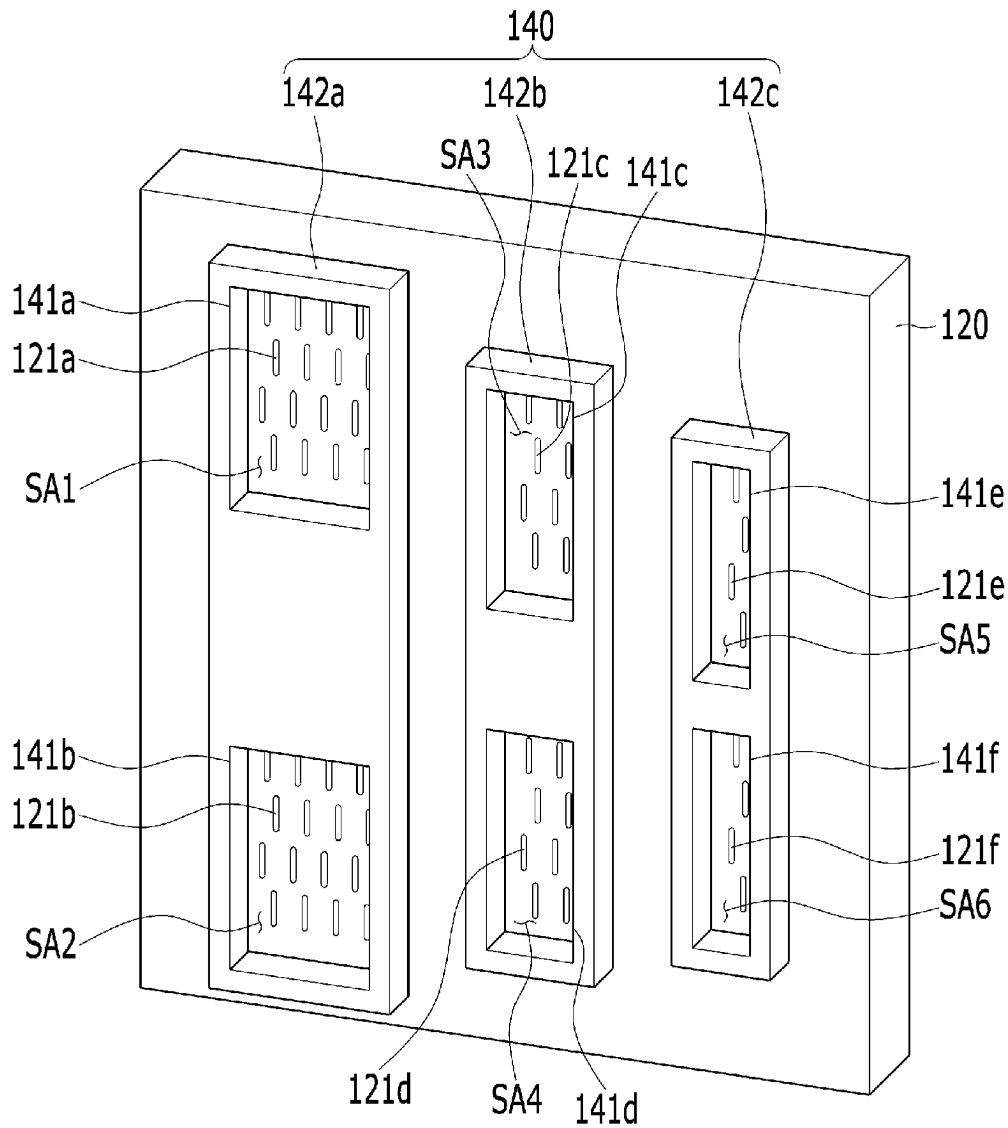
[FIG. 10]


141a
SA1
141c
SA3
121c
141e
SA5
121a
121e
121b
121f
141b
SA2
141d
SA4
121d
141f
SA6

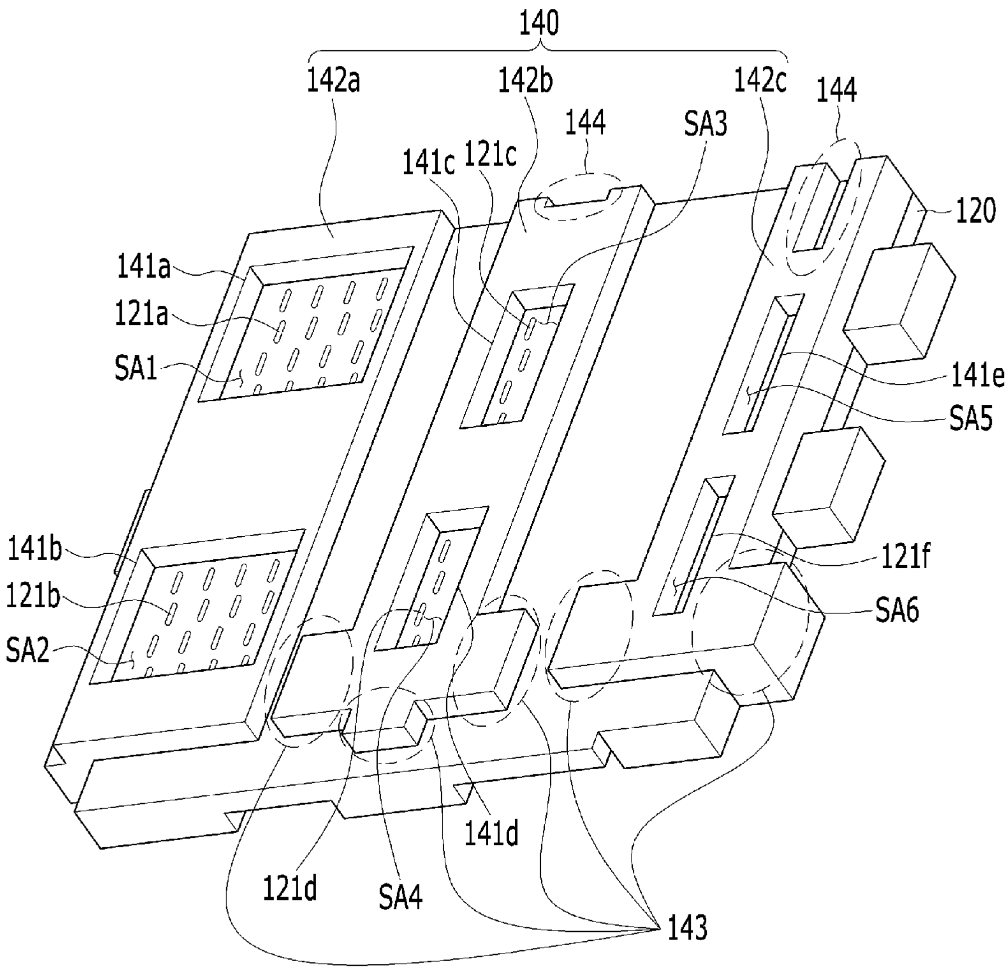
[FIG. 11]



[FIG. 12]

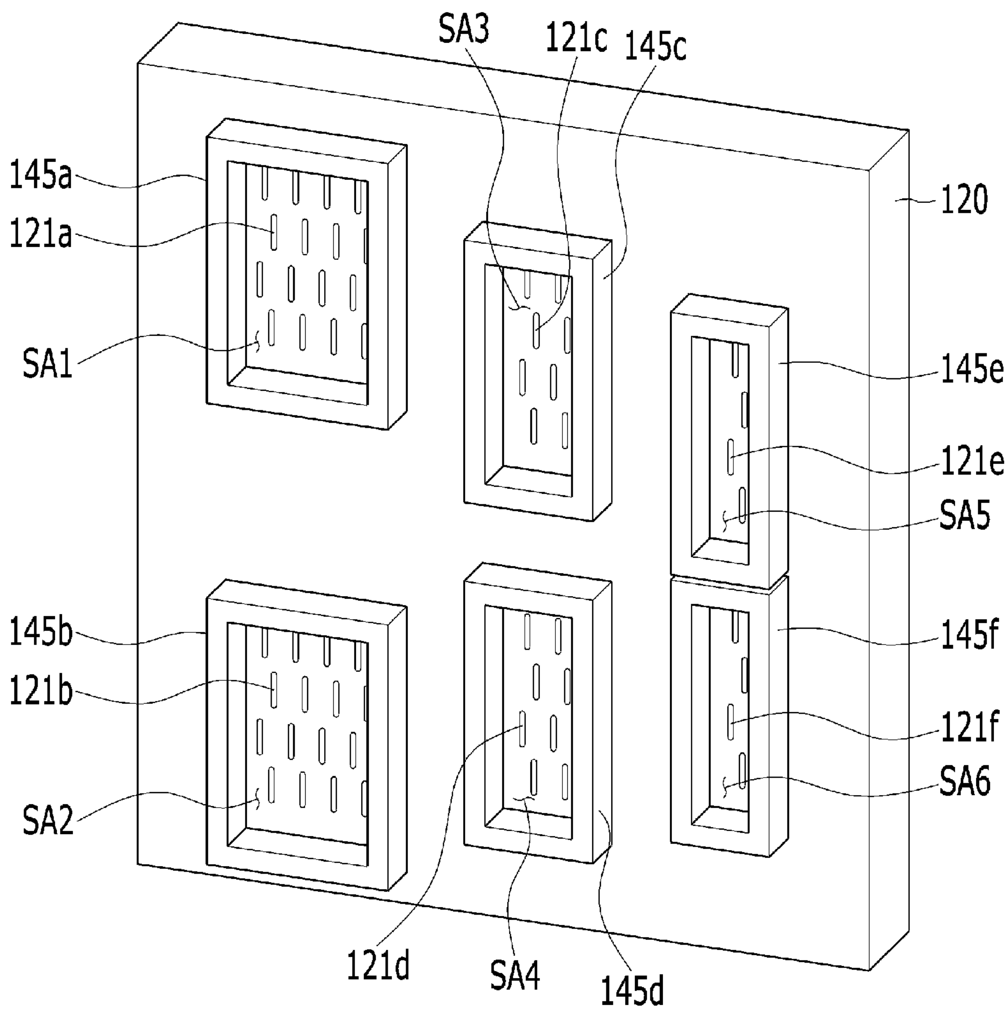

140
142a
142b
142c
SA3
121c
141c
141a
121a
SA1
120
141e
121e
SA5
141b
121b
SA2
141f
121f
SA6
121d
SA4
141d

[FIG. 13]

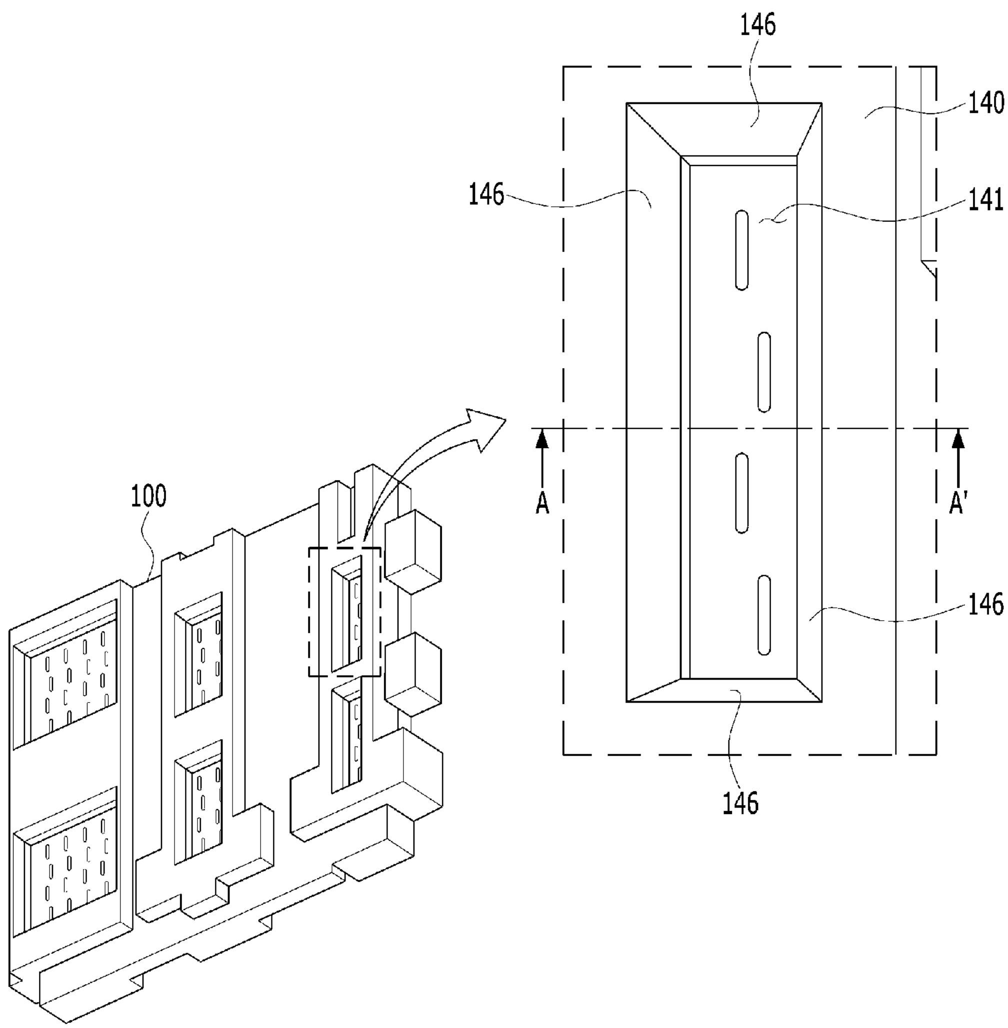


140
142a
142b
142c
144
144
SA3
141c
121c
120
141a
121a
SA1
141e
SA5
141b
121b
SA2
121f
SA6
141d
121d
SA4
143

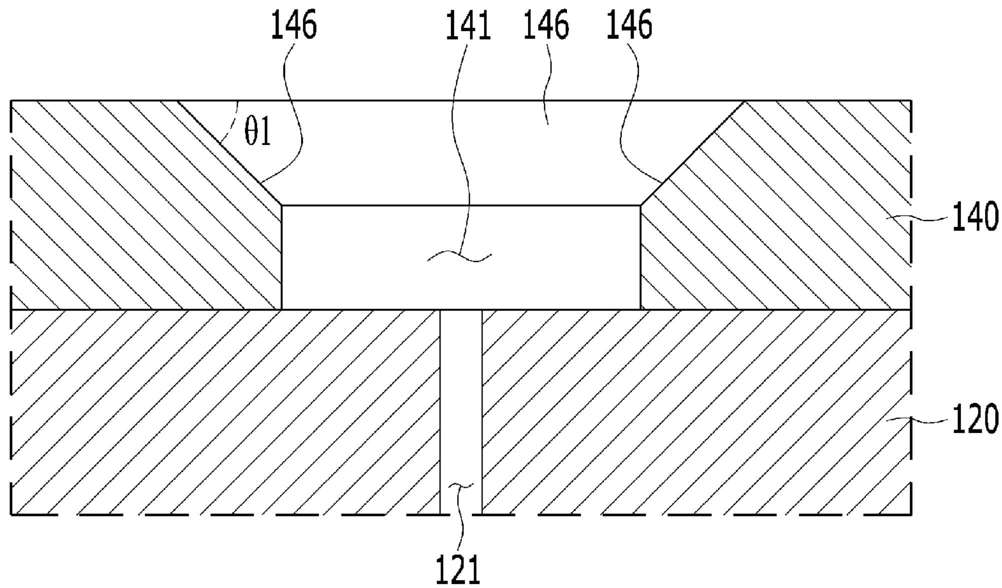
[FIG. 14]


SA3
121c
145c
145a
121a
SA1
120
145e
121e
SA5
145b
121b
SA2
145f
121f
SA6
121d
SA4
145d

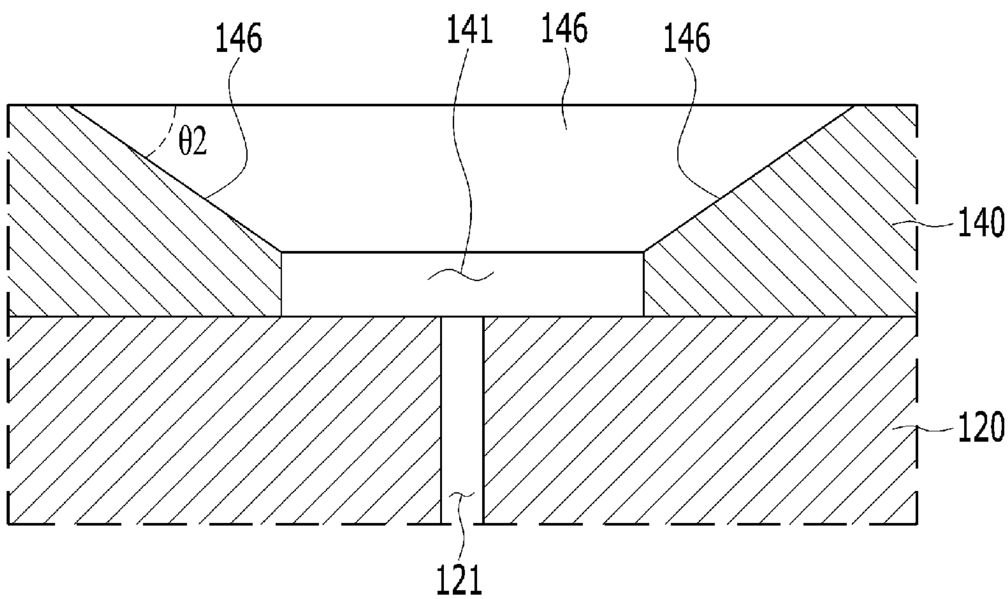
[FIG. 15]



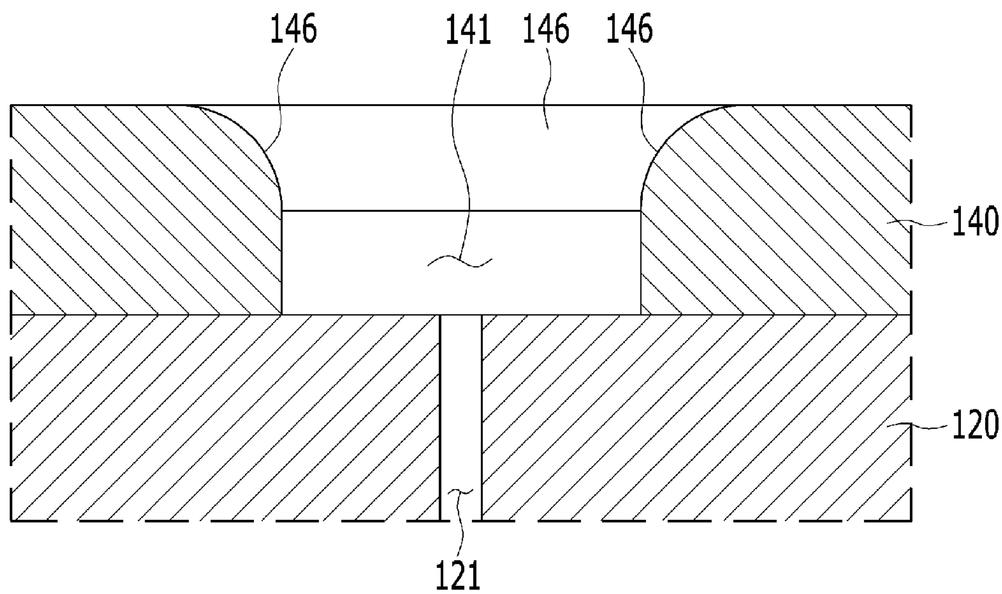
[FIG. 16]



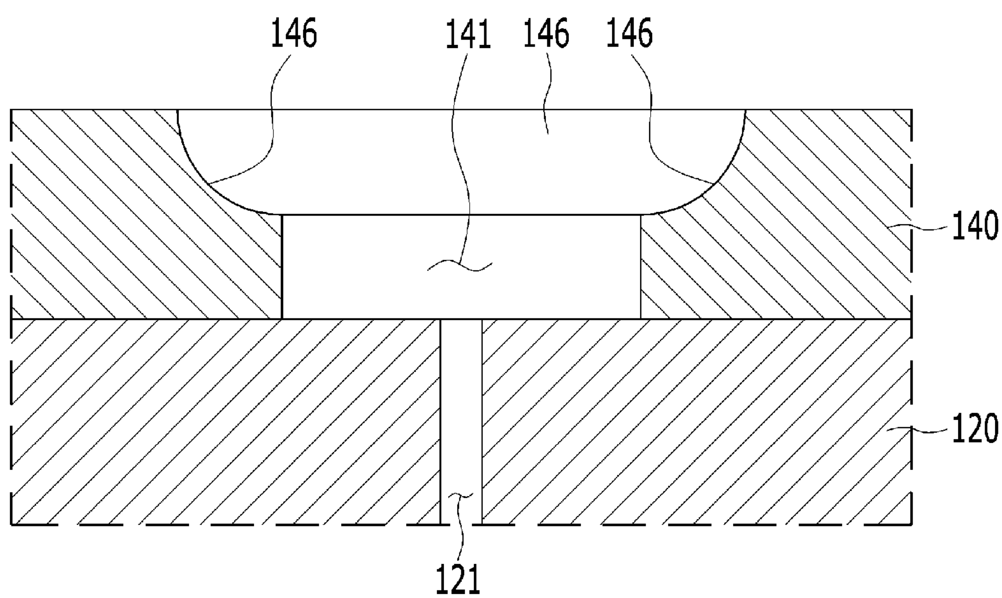
[FIG. 17]



[FIG. 18]



[FIG. 19]



RADAR ANTENNA

TECHNICAL FIELD

The present disclosure relates to an antenna, and more particularly, to a radar antenna.

BACKGROUND ART

It is on trend to use a radar antenna for signal transmission and reception for detecting an object around a vehicle. The radar antenna radiates radio waves onto an object, and makes it possible to detect the existence/nonexistence, distance, movement direction, movement speed, identification, and classification of the object by means of reflected waves or scattered waves having bounced off the object.

Recently, for an advancement of anti-collision radar of an autonomous vehicle to cope with a driverless vehicle era, technologies to widen the detection range and to heighten the performance of such a radar antenna have been researched.

In the radar antenna in the related art, since a plurality of radiators that play different roles are arranged, mutual coupling occurs among the radiators to cause independent performances of the radiators to deteriorate or to cause the radiation performance of the antenna to deteriorate.

The above matter described as a background technology is to help understanding of the background of the present disclosure, and may include the matter that is not the technology in the related art already known to those of ordinary skill in the art to which the present disclosure pertains.

DISCLOSURE

Technical Problem

The present disclosure has been proposed to solve the above-described problems, and an object of the present disclosure is to provide a radar antenna, which can prevent mutual coupling between antennas by forming a shielding space corresponding to each antenna of an antenna body by using accommodation holes of shielding members.

Technical Solution

In order to achieve that above object, a radar antenna according to an embodiment of the present disclosure includes: an antenna body which has a first surface and a second surface and in which a plurality of first slot groups are formed to be spaced apart from one another on the first surface; and a shielding member which is stacked on the first surface of the antenna body and in which a plurality of accommodation holes are formed to overlap the plurality of first slot groups, respectively.

The first slot group may include a plurality of first slots in a matrix arrangement, and the first surface of the antenna body may include a plurality of areas in which the plurality of first slots are disposed. In this case, the plurality of accommodation holes may overlap the plurality of areas in which the plurality of first slots are disposed.

The first surface of the antenna body and inner walls of the accommodation holes of the shielding member may form shielding spaces, and the inner walls of the accommodation holes may be spaced apart from outer circumference of areas occupied by the first slot group over a predetermined distance. In other words, the inner walls of the accommodation holes may be spaced apart from the outer circumference of the areas formed by the plurality of first slots included in the first slot group over the predetermined distance.

The shielding member may include a plurality of shielding plates stacked on the antenna body to be spaced apart from one another, two or more accommodation holes may be formed on the shielding plates of the shielding member, and the shielding plates of the shielding member may be stacked on the antenna body so as to overlap two or more of the plurality of first slot groups. In this case, a protrusion part extending in an outward direction or a groove part formed in an inward direction of the shielding plates may be formed on at least one of the plurality of shielding plates.

The shielding member may include a plurality of shielding blocks stacked on the antenna body to be spaced apart from one another, one accommodation hole may be formed on the shielding blocks of the shielding member, and the shielding blocks of the shielding member may be stacked on the antenna body so as to overlap one of the plurality of first slot groups.

The shielding member may include a chamfer part chamfered along a corner formed by the accommodation hole, and the chamfer part may be formed by chamfering the corner in a direction of the first surface of the shielding member.

The radar antenna according to an embodiment of the present disclosure may further include a slot member formed in a metal frame shape and configured to be inserted into the first slot formed on the antenna body.

Advantageous Effects

According to the present disclosure, the radar antenna has the effect of being able to prevent the mutual coupling among the antennas formed on the antenna body since the shielding spaces corresponding to the respective antennas of the antenna body are formed by using the accommodation holes of the shielding member. That is, according to the radar antenna, since the shielding member on which the accommodation holes are formed is stacked on the antenna body on which the plurality of antennas (slots) are formed, the antennas can be disposed in the shielding spaces formed by the accommodation holes and the antenna body, and thus the mutual coupling with other antennas can be prevented.

Further, since the radar antenna can prevent the mutual coupling among the antennas through stacking of the shielding member on the antenna body, the independent antenna performances of the plurality of antennas formed on the antenna body can be maintained, and thus the radiation performance of the radar antenna can be prevented from being degraded.

DESCRIPTION OF DRAWINGS

FIG. **1** is a perspective view of a radar antenna according to an embodiment of the present disclosure.

FIG. **2** is an exploded perspective view of a radar antenna when viewed in a direction of a first surface of the radar antenna according to an embodiment of the present disclosure.

FIG. **3** is an exploded perspective view of a radar antenna when viewed in a direction of a second surface that faces a first surface of the radar antenna according to an embodiment of the present disclosure.

FIG. **4** is a view explaining a first slot formed on the antenna body of FIG. **1**.

FIG. **5** is a view explaining a second slot formed on the antenna body of FIG. **1**.

FIGS. 6 and 7 are views explaining a waveguide formed on the antenna body of FIG. 1.

FIG. 8 is a view explaining a modified example of the antenna body of FIG. 1.

FIGS. 9 and 10 are views explaining the shielding member of FIG. 1.

FIG. 11 is a view explaining accommodation grooves of the shielding member of FIG. 9.

FIGS. 12 and 13 are views explaining a first modified example of the shielding member of FIG. 1.

FIG. 14 is a view explaining a second modified example of the shielding member of FIG. 1.

FIG. 15 is a view explaining an example of a chamfered surface formed in accommodation holes of the shielding member of FIG. 1.

FIGS. 16 to 19 are views explaining an example of the chamfered surface of FIG. formed in different shapes.

MODE FOR INVENTION

For detailed explanation to the extent that those of ordinary skill in the art to which the present disclosure pertains can easily embody the technical idea of the present disclosure, the most preferred embodiment of the present disclosure will be described with reference to the accompanying drawings. First, in giving reference numerals to constituent elements of the respective drawings, it is to be noted that the same constituent elements have possibly the same reference numerals although they are denoted in different drawings. Further, in describing the present disclosure, detailed explanation of related known constitutions or functions will be omitted in case that such detailed explanation may obscure the subject matter of the present disclosure.

Referring to FIGS. **1** to **3**, a radar antenna **100** according to an embodiment of the present disclosure is configured to include antennas and a shielding member **140**.

An antenna body **120** forms an appearance of the radar antenna **100**, and is formed in a flat plate shape having a predetermined thickness. The antenna body **120** has a first surface F**1** and a second surface F**2** facing the first surface F**1**. In this case, a board is disposed on the second surface F**2** of the antenna body **120**. The board **10** transmits radio waves to outside through the radar antenna **100**, and receives and performs signal processing of the radio waves received through the radar antenna **100**. Here, it is exemplified that the first surface F**1** is a front surface of the antenna body **120**, and the second surface F**2** is a rear surface of the antenna body **120**.

Referring to FIG. **4**, a plurality of first slots **121** are formed on the first surface F**1** of the antenna body **120**. The plurality of first slots **121** formed on the first surface F**1** of the antenna body **120** are classified into a plurality of first slot groups, and the plurality of first slot groups constitute respective antennas. In this case, the first slot group **121** constitutes one of an antenna for radiation and an antenna for reception. Here, the antenna for radiation is an antenna that radiates radio waves, and the antenna for reception is an antenna that receives the radio waves bounced and reflected from an object hit by the radio waves radiated from the antenna for radiation.

As an example, referring to FIG. **4**, on the first surface F**1** of the antenna body **120**, a plurality of the (1-1)-th slots **121***a*, a plurality of the (1-2)-th slots **121***b*, a plurality of the (1-3)-th slots **121***c*, a plurality of the (1-4)-th slots **121***d*, a plurality of the (1-5)-th slots **121***e*, and a plurality of the (1-6)-th slots **121***f* are formed, and 6 antennas are formed.

The plurality of the (1-1)-th slots **121***a* constitute a first antenna through matrix arrangement in a first area A**1**. The first area A**1** is a partial area of the first surface F**1** of the antenna body **120**, and is an area adjacent to a first side surface S**1** and a fourth side surface S**4** of the antenna body **120**.

The plurality of the (1-2)-th slots **121***b* constitute a second antenna through matrix arrangement in a second area A**2**. The second area A**2** is a partial area of the first surface F**1** of the antenna body **120**, and is an area adjacent to a third side surface S**3** and spaced apart for a predetermined distance from the first area A**1** downward.

The plurality of the (1-3)-th slots **121***c* constitute a third antenna through matrix arrangement in a third area A**3**. The third area A**3** is a partial area of the first surface F**1** of the antenna body **120**, and is an area adjacent to a first side surface S**1** and spaced apart for a predetermined distance from the first area A**1** rightward.

The plurality of the (1-4)-th slots **121***d* constitute a fourth antenna through matrix arrangement in a fourth area A**4**. The fourth area A**4** is a partial area of the first surface F**1** of the antenna body **120**, and is an area adjacent to the third side surface S**3**, spaced apart for a predetermined distance from the second area A**2** rightward, and spaced apart for a predetermined distance from the third area A**3** downward.

The plurality of the (1-5)-th slots **121***e* constitute a fifth antenna through matrix arrangement in a fifth area A**5**. The fifth area A**5** is a partial area of the first surface F**1** of the antenna body **120**, and is an area adjacent to the first side surface S**1** and the second side surface S**2**, and spaced apart for a predetermined distance from the third area A**3** rightward.

The plurality of the (1-6)-th slots **121***f* constitute a sixth antenna through matrix arrangement in a sixth area A**6**. The sixth area A**6** is a partial area of the first surface F**1** of the antenna body **120**, and is an area adjacent to the second side surface S**2** and the third side surface S**3**, spaced apart for a predetermined distance from the fourth area A**4** rightward, and spaced apart for a predetermined distance from the fifth area A**5** downward.

In this case, the first to sixth antennas operate as radiation antennas radiating the radio waves or reception antennas receiving the radio waves bounced and reflected from the object hit by the radio waves radiated from the radiation antenna.

Here, although it is exemplified that 6 antennas are formed in order to explain the antenna body **120** in FIG. **4**, the number of antennas may be changed depending on the required antenna characteristic, and thus is not limited thereto.

A plurality of second slots **122** are formed on the second surface F**2** of the antenna body **120**. The plurality of second slots **122** constitute a reception port and a transmission port. That is, the plurality of second slots **122** are configured as the transmission port connected to a transmission terminal of the board **10**, or are configured as the reception port that is connected to a reception terminal of the board **10**.

The plurality of second slots **122** are configured to match the antennas formed on the first surface F**1** of the antenna body **120**. One or more second slots **122** match the antennas formed on the first surface F**1** of the antenna body **120**. Accordingly, the number of second slots **122** is equal to or larger than the number of antennas formed on the first surface F**1** of the antenna body **120**.

As an example, referring to FIG. **5**, on the second surface F**2** of the antenna body **120**, the (2-1)-th slot **122***a*, the

(2-2)-th slot **122***b*, the (2-3)-th slot **122***c*, the (2-4)-th slot **122***d*, the (2-5)-th slot **122***e*, and the (2-6)-th slot **122***f* are formed.

The (2-1)-th slot **122***a* is formed on the second surface F**2** of the antenna body **120**, but is formed to overlap the first area A**1** on the first surface F**1** of the antenna body **120**.

The (2-2)-th slot **122***b* is formed on the second surface F**2** of the antenna body **120**, but is formed to overlap the second area A**2** on the first surface F**1** of the antenna body **120**.

The (2-3)-th slot **122***c* is formed on the second surface F**2** of the antenna body **120**, but is formed to overlap the third area A**3** on the first surface F**1** of the antenna body **120**.

The (2-4)-th slot **122***d* is formed on the second surface F**2** of the antenna body **120**, but is formed to overlap the fourth area A**4** on the first surface F**1** of the antenna body **120**.

The (2-5)-th slot **122***e* is formed on the second surface F**2** of the antenna body **120**, but is formed to overlap the fifth area A**5** on the first surface F**1** of the antenna body **120**.

The (2-6)-th slot **122***f* is formed on the second surface F**2** of the antenna body **120**, but is formed to overlap the sixth area A**6** on the first surface F**1** of the antenna body **120**.

A plurality of waveguides **123** that are paths for moving the radio waves are formed inside the antenna body **120**. The waveguides **123** form radio wave movement paths which connect the plurality of first slots **121** and the plurality of second slots **122** to each other, and move the radio waves between the first slots **121** and the second slots **122**.

As an example, referring to FIGS. **6** and **7**, the waveguide **123** is formed inside the antenna body **120**, and forms the radio wave movement path for moving (radiating) the radio waves output from the board **10** to an outside of the radar antenna **100** or moving the radio waves reflected from an object to the board **10** (i.e., reception terminal).

Meanwhile, since the radar antenna is an antenna composed of a plurality of slots, the antenna performance is sensitively changed depending on the dimensions (size) of the slots in a high frequency band of about 76.5 GHz.

In case that the radar antenna is produced of plastic to lower the product unit price (production unit price), the radar antenna has the problems in that it is difficult to precisely process the dimensions (size) of the slots due to the production tolerance, and it is difficult to implement the constant antenna performance due to the dimension change of the slots.

Accordingly, referring to FIG. **8**, the radar antenna **100** according to an embodiment of the present disclosure may further include a plurality of slot members **150** inserted into the first slots **121**.

The slot member **150** is formed of a metal material, and is formed in the shape of a frame in which the slot is formed. The slot member **150** is fixedly inserted into the first slot **121** of the antenna body **120**. For this, the slot member **150** is formed to have the same dimensions as the design dimensions of the first slot **121**.

In the radar antenna **100**, since the slot member **150** is inserted in a manner of being fitted into the first slot **121**, the production tolerance can be reduced, and thus the dimensions of the first slot **121** can be constantly formed.

Further, in case that dimension inferiority of the first slot **121** occurs, the dimensions can be constantly maintained by replacing the slot member **150**, and thus the radar antenna **100** has an additional effect of heightening the yield.

Further, the radar antenna **100** can constantly implement the antenna performance by making the dimensions of the first slot **121** constant.

The above-described antenna body **120** may operate as the radar antenna **100** by itself. However, since the antenna body **120** includes a plurality of antennas which transmit/receive the frequency band to/from each other or operate for frequency transmission or reception, mutual coupling among the antennas may occur to deteriorate independent antenna performances.

Accordingly, the radar antenna **100** according to an embodiment of the present disclosure further includes the shielding member **140** stacked on the first surface F**1** of the antenna body **120** to prevent the mutual coupling among the antennas.

The shielding member **140** is stacked on the first surface F**1** of the antenna body **120** on which the plurality of first slots **121** are formed. The shielding member **140** may be formed of a metal that shields signals to prevent the mutual coupling between the antennas formed by the plurality of first slots **121**. However, the shielding member **140** may be replaced by another material other than the metal, or another metal depending on the antenna specifications required in the radar antenna **100**.

Further, the thickness of the shielding member **140** and the gap between the first slot **121** and the shielding member **140** may differ depending on the antenna characteristics, antenna sizes, and surrounding environments. Accordingly, in an embodiment of the present disclosure, the material, shape, thickness, and gap of the shielding member **140** are not limited to numbers.

Referring to FIGS. **9** and **10**, the shielding member **140** has a first surface F**3** and a second surface F**4**, and the second surface F**4** is stacked on the first surface F**1** of the antenna body **120**.

A plurality of accommodation holes **141** for accommodating the first slots **121** are formed on the shielding member **140**. The accommodation holes **141** accommodate some of the plurality of first slots **121**, but accommodate the plurality of first slots **121** belonging to the same slot group. The accommodation holes **141** are formed to penetrate the shielding member **140** in the shape of a flat plate, and are formed in the shape of a quadrangle when viewed in a direction of the first surface F**1** of the antenna body **120**. In this case, the accommodation holes **141** may be formed in another shape, such as in the shape of a circle, depending on the required antenna specifications.

The shielding member **140** forms shielding spaces for preventing the mutual coupling among the antennas formed by the plurality of first slots **121**. As the shielding member **140** is stacked on the antenna body **120**, the accommodation holes **141** and the first surface F**1** of the antenna body **120** form the shielding spaces. The inner walls of the accommodation holes **141** form shielding walls that are outer walls of the shielding spaces for shielding between the antennas formed by the first slots **121**, and the first surface F**1** of the antenna body **120** forms the bottom surface of the shielding spaces. Accordingly, the antenna body **120** and the shielding member **140** form the shielding spaces being open in the direction of the first surface F**3** of the shielding member **140**. Through this, the shielding member **140** can prevent the mutual coupling among the antennas being formed by the plurality of first slots **121**.

As an example, the plurality of accommodation holes **141** include a first accommodation hole **141***a*, a second accommodation hole **141***b*, a third accommodation hole **141***c*, a fourth accommodation hole **141***d*, a fifth accommodation hole **141***e*, and a sixth accommodation hole **141***f*.

The first accommodation hole **141***a* is formed in an area that overlaps the first area A**1** of the antenna body **120** on which a plurality of the (1-1)-th slots **121***a* are formed. As the shielding member **140** is stacked on the antenna body

120, the first accommodation hole **141***a* accommodates the plurality of the (1-1)-th slots **121***a*. On the inner wall of the first accommodation hole **141***a* and the first surface F**1** of the antenna body **120**, the first area A**1** forms first shielding spaces SA**1**, and since the first shielding spaces SA**1** are open in the direction of the first surface F**3** of the shielding member **140**, the first shielding spaces SA**1** do not exert an influence on the radiation and reception of the radio waves, and block the mutual coupling between the first antenna being formed by the plurality of the (1-1)-th slots **121***a* and other antennas.

The second accommodation hole **141***b* is formed in an area that overlaps the second area A**2** of the antenna body **120** on which a plurality of the (1-2)-th slots **121***b* are formed. As the shielding member **140** is stacked on the antenna body **120**, the second accommodation hole **141***b* accommodates the plurality of the (1-2)-th slots **121***b*. On the inner wall of the second accommodation hole **141***b* and the first surface F**1** of the antenna body **120**, the second area A**2** forms second shielding spaces SA**2**, and since the second shielding spaces SA**2** are open in the direction of the first surface F**3** of the shielding member **140**, the second shielding spaces SA**2** do not exert an influence on the radiation and reception of the radio waves, and block the mutual coupling between the second antenna being formed by the plurality of the (1-2)-th slots **121***b* and other antennas.

The third accommodation hole **141***c* is formed in an area that overlaps the third area A**3** of the antenna body **120** on which a plurality of the (1-3)-th slots **121***c* are formed. As the shielding member **140** is stacked on the antenna body **120**, the third accommodation hole **141***c* accommodates the plurality of the (1-3)-th slots **121***c*. On the inner wall of the third accommodation hole **141***c* and the first surface F**1** of the antenna body **120**, the third area A**3** forms third shielding spaces SA**3**, and since the third shielding spaces SA**3** are open in the direction of the first surface F**3** of the shielding member **140**, the third shielding spaces SA**3** do not exert an influence on the radiation and reception of the radio waves, and block the mutual coupling between the third antenna being formed by the plurality of the (1-3)-th slots **121***c* and other antennas.

The fourth accommodation hole **141***d* is formed in an area that overlaps the fourth area A**4** of the antenna body **120** on which a plurality of the (1-4)-th slots **121***d* are formed. As the shielding member **140** is stacked on the antenna body **120**, the fourth accommodation hole **141***d* accommodates the plurality of the (1-4)-th slots **121***d*. On the inner wall of the fourth accommodation hole **141***d* and the first surface F**1** of the antenna body **120**, the fourth area A**4** forms fourth shielding spaces SA**4**, and since the fourth shielding spaces SA**4** are open in the direction of the first surface F**3** of the shielding member **140**, the fourth shielding spaces SA**4** do not exert an influence on the radiation and reception of the radio waves, and block the mutual coupling between the fourth antenna being formed by the plurality of the (1-4)-th slots **121***d* and other antennas.

The fifth accommodation hole **141***e* is formed in an area that overlaps the fifth area A**5** of the antenna body **120** on which a plurality of the (1-5)-th slots **121***e* are formed. As the shielding member **140** is stacked on the antenna body **120**, the fifth accommodation hole **141***e* accommodates the plurality of the (1-5)-th slots **121***e*. On the inner wall of the fifth accommodation hole **141***e* and the first surface F**1** of the antenna body **120**, the fifth area A**5** forms fifth shielding spaces SA**5**, and since the fifth shielding spaces SA**5** are open in the direction of the first surface F**3** of the shielding member **140**, the fifth shielding spaces SA**5** do not exert an influence on the radiation and reception of the radio waves, and block the mutual coupling between the fifth antenna being formed by the plurality of the (1-5)-th slots **121***e* and other antennas.

The sixth accommodation hole **141***f* is formed in an area that overlaps the sixth area A**6** of the antenna body **120** on which a plurality of the (1-6)-th slots **121***f* are formed. As the shielding member **140** is stacked on the antenna body **120**, the sixth accommodation hole **141***f* accommodates the plurality of the (1-6)-th slots **121***f*. On the inner wall of the sixth accommodation hole **141***f* and the first surface F**1** of the antenna body **120**, the sixth area A**6** forms fifth shielding spaces SA**6**, and since the sixth shielding spaces SA**6** are open in the direction of the first surface F**3** of the shielding member **140**, the sixth shielding spaces SA**6** do not exert an influence on the radiation and reception of the radio waves, and block the mutual coupling between the sixth antenna being formed by the plurality of the (1-6)-th slots **121***f* and other antennas.

Meanwhile, if the inner walls of the accommodation holes and the outer circumference of the area in which the first slots **121** are formed are spaced apart from each other below a predetermined gap, the radiation or reception performance of the radio waves may deteriorate.

Accordingly, referring to FIG. **11**, the inner walls of the accommodation holes **141** are disposed to be spaced apart from the first slots **121** formed on the first surface F**1** of the antenna body **120** over a predetermined gap d. That is, the inner walls of the accommodation holes **141** are disposed to be spaced apart from the outer circumference of the area in which the first slots **121** are formed over the predetermined gap (predetermined distance) d. Here, the predetermined gap d may differ depending on the size of the radar antenna **100** and the antenna characteristics, and is not limited to numbers.

Referring to FIG. **12**, the shielding member **140** may include a plurality of shielding plates **142**. That is, the shielding member **140** may include a first shielding plate **142***a*, a second shielding plate **142***b*, and a third shielding plate **142***c*.

The first shielding plate **142***a* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the first area A**1** and the second area A**2** of the antenna body **120**. The first accommodation hole **141***a* and the second accommodation hole **141***b* are formed on the first shielding plate **142***a*, and the first shielding plate **142***a* forms the first shielding space SA**1** and the second shielding space SA**2** as being stacked on the antenna body **120**.

The second shielding plate **142***b* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the third area A**3** and the fourth area A**4** of the antenna body **120**. The third accommodation hole **141***c* and the fourth accommodation hole **141***d* are formed on the second shielding plate **142***b*, and the second shielding plate **142***b* forms the third shielding space SA**3** and the fourth shielding space SA**4** as being stacked on the antenna body **120**.

The third shielding plate **142***c* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the fourth area A**4** and the fifth area A**5** of the antenna body **120**. The fifth accommodation hole **141***e* and the sixth accommodation hole **141***f* are formed on the third shielding plate **142***c*, and the third shielding plate **142***c* forms the fifth shielding space SA**5** and the sixth shielding space SA**6** as being stacked on the antenna body **120**.

In this case, referring to FIG. **13**, a protrusion part **143** or a groove part **144** may be further formed on the first shielding plate **142***a* to the third shielding plate **142***c* in accordance with the antenna specifications. Here, it is exemplified that the protrusion part **143** is an area extending in an outward direction (or lateral direction) on the shielding plate **142**, and the groove part **144** is an area formed in an inward direction of the shielding plate **142** through removal of a part of the shielding plate **142**.

Referring to FIG. **14**, the shielding member may be composed of a plurality of shielding blocks **145**.

As an example, the shielding member **140** may include a first shielding block **145***a*, a second shielding block **145***b*, a third shielding block **145***c*, a fourth shielding block **145***d*, a fifth shielding block **145***e*, and a sixth shielding block **145***f*.

The first shielding block **145***a* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the first area A**1** of the antenna body **120**. The first accommodation hole **141***a* is formed on the first shielding block **145***a*, and the first shielding block **145***a* forms the first shielding space SA**1** as being stacked on the antenna body **120**.

The second shielding block **145***b* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the second area A**2** of the antenna body **120**. The second accommodation hole **141***b* is formed on the second shielding block **145***b*, and the second shielding block **145***b* forms the second shielding space SA**2** as being stacked on the antenna body **120**.

The third shielding block **145***c* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the third area A**3** of the antenna body **120**. The third accommodation hole **141***c* is formed on the third shielding block **145***c*, and the third shielding block **145***c* forms the third shielding space SA**3** as being stacked on the antenna body **120**.

The fourth shielding block **145***d* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the fourth area A**4** of the antenna body **120**. The fourth accommodation hole **141***d* is formed on the fourth shielding block **145***d*, and the fourth shielding block **145***d* forms the fourth shielding space SA**4** as being stacked on the antenna body **120**.

The fifth shielding block **145***e* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the fifth area A**5** of the antenna body **120**. The fifth accommodation hole **141***e* is formed on the fifth shielding block **145***e*, and the fifth shielding block **145***e* forms the fifth shielding space SA**5** as being stacked on the antenna body **120**.

The sixth shielding block **145***f* is stacked on the first surface F**1** of the antenna body **120**, but is stacked to overlap the sixth area A**6** of the antenna body **120**. The sixth accommodation hole **141***f* is formed on the sixth shielding block **145***f*, and the sixth shielding block **145***f* forms the sixth shielding space SA**6** as being stacked on the antenna body **120**.

Referring to FIG. **15**, in order to improve a gain of the radar antenna **100**, a chamfer part **146** may be formed on the shielding member **140**. The chamfer part **146** is formed by chamfering the corner of the area in which the accommodation holes **141** are formed on the shielding member **140**. The chamfer part **146** is formed along the upper corner of the shielding member **140**.

In this case, the chamfer part **146** may be formed in various shapes in accordance with the antenna characteristics.

As an example, referring to FIG. **16**, the chamfer part **146** is formed in a diagonal shape based on a cutting plane obtained by cutting the shielding member **140** based on line A-A', but is chamfered to form a predetermined angle θ**1** with the upper surface of the shielding member **140**.

In this case, referring to FIG. **17**, the chamfer part **146** is formed in a diagonal shape based on a cutting plane obtained by cutting the shielding member **140** based on line A-A', but is chamfered to form another predetermined angle θ**2** with the upper surface of the shielding member **140**.

As another example, referring to FIG. **18**, the chamfer part **146** may be formed in the shape of an annular surface obtained by chamfering round the corners based on the cutting plane obtained by cutting the shielding member **140** based on line A-A'.

As another example, referring to FIG. **19**, the chamfer part **146** may be formed in the shape of a concave surface obtained by concavely chamfering the corners based on the cutting plane obtained by cutting the shielding member **140** based on line A-A'.

Here, although FIGS. **16** to **19** are exemplarily illustrated for easy explanation of the chamfer part **146**, the chamfer part **146** is not limited thereto, and may be modified in various shapes in accordance with the characteristics and the purpose of the antenna.

As described above, although a preferred embodiment according to the present disclosure has been described, it is understood that various modifications are possible, and those of ordinary skill in the corresponding technical field can make various modifications and correction examples without deviating from the claims of the present disclosure.

The invention claimed is:

1. A radar antenna comprising:
- an antenna body which has a first surface and a second surface and in which a plurality of first slot groups are formed to be spaced apart from one another on the first surface;
- a shielding member which is stacked on the first surface of the antenna body and in which a plurality of accommodation holes are formed to overlap the plurality of first slot groups, respectively; and
- a slot member formed in a metal frame shape and configured to be inserted into a first slot formed on the antenna body.

2. The radar antenna of claim **1**, wherein each first slot group of the plurality of first slot groups comprises a plurality of first slots in a matrix arrangement.

3. The radar antenna of claim **1**, wherein the first surface of the antenna body comprises a plurality of areas in which a plurality of first slots are disposed.

4. The radar antenna of claim **3**, wherein the plurality of accommodation holes overlap the plurality of areas in which the plurality of first slots are disposed.

5. The radar antenna of claim **1**, wherein the first surface of the antenna body and inner walls of the accommodation holes of the shielding member form shielding spaces.

6. The radar antenna of claim **1**, wherein the inner walls of the accommodation holes are spaced apart from outer peripheries of areas occupied by a first slot group over a predetermined distance.

7. The radar antenna of claim **1**, wherein inner walls of the accommodation holes are spaced apart from the outer peripheries of the areas formed by a plurality of first slots included in a first slot group over the predetermined distance.

8. The radar antenna of claim **1**, wherein the shielding member comprises a plurality of shielding plates stacked on the antenna body to be spaced apart from one another,
- two or more accommodation holes are formed on a shielding plate of the shielding member, and

the shielding plate of the shielding member is stacked on the antenna body so as to overlap two or more first slot groups of the plurality of first slot groups.

9. The radar antenna of claim **8**, wherein a protrusion part extending in an outward direction is formed on at least one of the plurality of shielding plates.

10. The radar antenna of claim **8**, wherein a part of at least one of the plurality of shielding plates is removed, and a groove part formed in an inward direction of the shielding plates is formed thereon.

11. The radar antenna of claim **1**, wherein the shielding member comprises a plurality of shielding blocks stacked on the antenna body to be spaced apart from one another,

one accommodation hole is formed on a shielding block of the shielding member, and

the shielding block of the shielding member is stacked on the antenna body so as to overlap one first slot group of the plurality of first slot groups.

12. The radar antenna of claim **1**, wherein the shielding member comprises a chamfer part chamfered along a corner formed by an accommodation hole.

13. The radar antenna of claim **12**, wherein the chamfer part is formed by chamfering the corner in a direction of the first surface of the shielding member.

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